

Indian Institute of Technology Patna
April 2025
Department of Computer Science and Engineering
Endsem Exam CS2201: *Formal Language and Automata Theory*

Duration: 3 hours

Full Marks = 100

Instructions:

- Sections I and II should be answered in the main copy provided (not in the question paper).
- Select all correct answers for multiple-choice questions (MCQs). Partial correct answers receive no marks. If "All of the above" or "None of the above" is correct, mark only that option; otherwise, no marks will be awarded.
- There is no negative marking
- No doubts will be entertained.

Section-I

Full Marks: 60

1. Savitch's Theorem implies which of the following? [1 mark]
 - (a) $P = NP$
 - (b) $NL = L$
 - (c) $NP = PSPACE$
 - (d) $NSPACE(S(n)) \subseteq SPACE(S(n)^2)$
2. Which language is accepted by the Turing Machine with input alphabet $\Sigma = \{a\}$ and with a transition having the head keeps moving right on 'a' and accepts on blank? [1 mark]
 - (a) $\{a^n b^n \mid n \geq 1\}$
 - (b) $\{a^*\}$
 - (c) $\{b^*\}$
 - (d) $\{a^n b^m \mid n, m \geq 0\}$
3. Which of the following machine is NOT proven to have the same computational power as the standard Turing machine? [1 marks]
 - (a) Stay-Option Turing Machine
 - (b) Multi-tape Turing Machine
 - (c) Semi-infinite Tape Turing Machine
 - (d) Pushdown Automaton
4. Which of the following statements is/are true? [2 marks]
 - (a) RE languages are closed under union, intersection and complementation.
 - (b) RE languages are closed under union and intersection but not under complementation.
 - (c) RE languages are closed under union and complementation but not under intersection.
 - (d) RE languages are closed under neither of union, intersection and complementation.
5. A 2-tape NTM is equivalent to: [2 marks]
 - (a) A 1-tape DTM
 - (b) A PDA
 - (c) An NFA
 - (d) None of the machines in the other options
6. Which of the following is/are NOT true for Turing Machines? [2 marks]
 - (a) They can have partial transition functions.
 - (b) They can compute any computable function.
 - (c) They can simulate any finite automaton.
 - (d) Turing Machines always halt for every input and have an infinite tape.
7. The set of all Turing machines is _____, while the set of all languages over a finite alphabet is _____. [2 marks]
8. In the reduction from 3CNF-SAT to CLIQUE (as per the class notes), how many nodes are in the clique if the formula has k clauses? [2 marks]
 - (a) k
 - (b) $2k$
 - (c) $3k$
 - (d) k^2
9. Which of the following is true about NP-complete problems? [2 marks]
 - (a) All NP-complete problems are in P.
 - (b) If any NP-complete problem is in P, then $P = NP$.
 - (c) All NP-complete problems are undecidable.
 - (d) NP-complete problems are not in NP.
10. What is the main idea behind polynomial-time reduction? [2 marks]
 - (a) To convert a problem into another in exponential time
 - (b) To show one problem is harder than another
 - (c) To solve a problem using a solution to another problem in polynomial time
 - (d) To prove problems are undecidable
11. What is the main restriction of a Linear Bounded Automaton compared to a Turing Machine? [2 marks]
 - (a) Number of states
 - (b) Tape alphabet
 - (c) Tape space is limited to input length
 - (d) No restriction
12. In the Chomsky Hierarchy, unrestricted grammars generate all _____ language. [2 marks]

13. The reduction from problem A to problem B is used to show: [2 marks]
- A is regular
 - B is regular
 - The decidability or undecidability of A and B
 - The time complexity of A
14. To check if a regular language accepted by DFA is infinite, one should: [2 marks]
- Check for cycles on a path from initial to accepting state
 - Count the number of accepting states
 - Check if the DFA has more than one state
 - Use the pumping lemma directly
15. The complement of a decidable language is: [2 marks]
- Always undecidable
 - Always decidable
 - Sometimes undecidable
 - Never Turing-Acceptable
16. Which of the following statements is/are TRUE? [2 marks]
- Every subset of a recursively enumerable language is recursive.
 - If a language L and its complement $\bar{L}$ are both recursively enumerable, then L must be recursive.
 - Complement of a context-free language must be recursive.
 - If L_1 and L_2 are regular, then $L_1 \cap L_2$ must be deterministic context-free.
17. Let X be a recursive language and Y be a recursively enumerable but not recursive language. Let W and Z be two languages such that $\bar{Y}$ reduces to W , and Z reduces to $\bar{X}$. Which one of the following statements is TRUE? [3 marks]
- W can be recursively enumerable and Z is recursive.
 - W can be recursive and Z is recursively enumerable.
 - W is not recursively enumerable and Z is recursive.
 - W is not recursively enumerable and Z is not recursive.
18. Which of the following decision problems are undecidable? [3 marks]
- Given NFAs N_1 and N_2 , is $L(N_1) \cap L(N_2) = \emptyset$?
 - Given a CFG $G = (N, \Sigma, P, S)$ and a string $x \in \Sigma^*$, does $x \in L(G)$?
 - Given CFGs G_1 and G_2 , is $L(G_1) = L(G_2)$?
 - Given a TM M , is $L(M) = \emptyset$?
- Options:**
- III and IV only
 - I and IV only
 - II and III only
 - II and IV only
19. Which of the following statements is/are TRUE? [3 marks]
- The intersection of two regular languages is regular.
 - The intersection of two context-free languages is context-free.
 - The intersection of two recursive languages is recursive.
 - The intersection of two recursively enumerable languages is recursively enumerable.
20. Consider the language
- $$L = \{M\#N\#x \mid M, N \text{ are Turing machines over } \Sigma, x \in \Sigma^*, \text{ and both } M \text{ and } N \text{ accept } x\}$$
- Which of the following statements is/are true? [3 marks]
- L is recursively enumerable.
 - L is not recursively enumerable.
 - If $x \in \mathcal{L}(M) \cup \mathcal{L}(N)$, then $M\#N\#x \in L$.
 - If $x \in \mathcal{L}(M)$ but $x \notin \mathcal{L}(N)$, then $M\#N\#x \in L$.
21. What language is generated by the unrestricted grammar $G = (\{S, B, a, b, c\}, \{a, b, c\}, R, S)$, where R consists of the following productions? [3 marks]
- $$\begin{aligned} S &\rightarrow aBSccc \mid aBccc \\ Ba &\rightarrow aB \\ Bc &\rightarrow bbc \\ Bb &\rightarrow bbb \end{aligned}$$
- $\{a^n b^{3n} c^{3n} \mid n \geq 0\}$
 - $\{a^n b^{2n} c^{3n} \mid n \geq 0\}$
 - $\{a^n b^n c^n \mid n > 0\}$
 - $\{a^n b^{2n} c^{3n} \mid n > 0\}$
22. Consider the hierarchy of languages shown in the Figure 1 below. Place each of the following languages at the proper place in the hierarchy, that is, deduce whether for each $i = 1, \dots, 4$ the language L_i as defined below is 'finite' or 'regular-and-infinite' or 'context-free-but-not-regular' or 'recursive-but-non-context-free' or 'R.E.-but-not-recursive' or 'non-R.E.'. You should provide sufficient justifications in brief for your judgments. [4*4 = 16 marks]
- The set L_1 of all strings $\alpha \in \{a, b, c\}^*$ containing an equal number of occurrences of a 's, b 's and c 's.
 - The complement of L_1 , that is, $L_2 := \{a, b, c\}^* \setminus L_1$.
 - $L_3 := \mathcal{L}(G)$, where G is the grammar
- $$(\{a, b\}, \{S, T\}, S, \{S \rightarrow \epsilon \mid abTS, T \rightarrow \epsilon \mid Tb\}).$$
- $L_4 := \{\langle M_i \rangle \mid \epsilon \notin \mathcal{L}(M_i)\}$, where $\langle M_i \rangle$ is the code for the i -th Turing machine with the binary input alphabet.

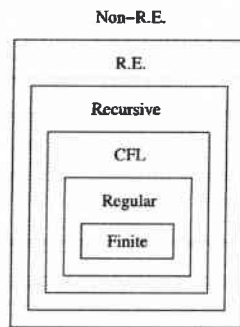


Figure 1:

Section-II

Full Marks:40

1. Design (Formal description and key idea as a step by step algorithm, do not need to draw the diagram) a Turing Machine to accept the language $L = \{a^n b^m a^{(n+m)} \mid n, m \geq 1\}$. Describe the steps it performs on input string 'aabbbaaaaa' and 'aabaaaa' [6+4 marks]
2. Suppose you are given a 2-tape Turing machine M that recognizes the language $L = \{a^n b^n \mid n \geq 1\}$ in $O(n)$ time by copying the b 's to the second tape and comparing with the a 's on the first tape. [4+4 marks]
 - (a) Explain how a standard single-tape Turing machine can simulate this 2-tape machine.
 - (b) What is the time complexity of the simulation, and why?
3. Prove that the language CONN (graph connectivity) is in NL. [4 marks]
4. Explain and prove that there exists a language that is not accepted by any Turing Machine. [9 marks]
5. Explain, with a detailed construction, how a 3CNF-SAT problem can be reduced to a CLIQUE problem. Illustrate the reduction with a specific example formula. [9 marks]

